

Ensuring 24x7 Infrastructure Resilience and End-User Reliability for a Leading U.S. Healthcare Organization

Introduction

A U.S.-based healthcare enterprise specializing in home-centered, whole-person care coordination partnered with NuWare to enhance the reliability, scalability, and performance of its IT infrastructure. The client's business model emphasizes value-based healthcare delivery, where data-driven insights enable timely, personalized care transitions from hospital to home.

With rapidly expanding operations and an increasing dependency on digital systems, the company required comprehensive 24x7 Infrastructure Monitoring (NOC) and End-User Helpdesk Support. NuWare, serving as the client's technology partner since 2019, implemented a multi-layered monitoring and support model to ensure operational continuity, efficient issue resolution, and SLA adherence across all systems.

The Challenge

The client's distributed infrastructure presented multiple technical and operational challenges:

1. **Complex Multi-Tier Infrastructure**

The environment included on-premise and cloud-based servers across Windows, Linux, and AS400 systems, supporting a variety of business-critical healthcare applications.

2. **24x7 Availability Requirements**

With a large user base and critical data operations, even minimal downtime impacted care delivery and patient management.

3. **Multiple Monitoring Tools and Integrations**

Server, application, network, and telephony monitoring required integration across disparate tools such as SolarWinds, AppDynamics, OEM, and AWS CloudWatch.

4. **SLA-Driven Response Expectations**

Required 10-minute response time for Priority 1 (P1) alerts and continuous coordination across teams to resolve major incidents.

5. **Operational Scalability and Compliance**

The client sought a flexible model capable of handling growing infrastructure assets while adhering to healthcare compliance and audit standards.

NuWare was chosen among several solution providers for its proven expertise in infrastructure monitoring, automation, and healthcare IT governance.

NuWare's Approach

NuWare deployed a 24x7x365 offshore Network Operations Center (NOC) and End-User Helpdesk Support Team, integrating proactive monitoring, SLA-based alert management, and automated workflows for faster issue resolution.

Engagement Overview

Parameter	Details
Client	U.S.-Based Healthcare Company
Engagement Start	2019
Team Size	15
Support Model	24x7x365
Environment	On-Premise + AWS Cloud
Monitoring Tools	SolarWinds, AppDynamics, OEM, Tidal, Cisco UCS, Cloudera, Airflow, AWS CloudWatch
Other Tools	Jira, Jenkins, Confluence, Bitbucket, Ansible
Support Functions	NOC Support, Helpdesk Support, L1 Application Support Case study - NOC and Helpdesk s...

NOC Support

As the **first line of defense**, NuWare's NOC team provides continuous monitoring and proactive maintenance across infrastructure, applications, databases, and networks.

Key Capabilities:

1. 24x7 Infrastructure Monitoring

- Continuous server, application, and database health checks.
- Monitoring network bandwidth, VOIP systems, and application uptime.
- Automated **alert escalation** and SLA-based tracking for all incidents.

2. Monitoring Configuration & Optimization

- Added/removed nodes and URLs in **SolarWinds** and **AppDynamics**.
- Configured dashboards to track **Server/Application/DB/Network/VOIP** metrics.

- c. Defined **business health transactions** and set **threshold limits** for real-time alerts.

3. Incident Management

- a. Reviewed and triaged incidents in **Jira**; escalated to relevant teams.
- b. Maintained a clean incident queue and tracked root cause analysis (RCA).
- c. Managed **Crisis Bridge Calls** for P1 issues, coordinating between teams and maintaining consistent outage communication.

4. Application Patching & Releases

- a. Managed application patch deployments via **Jenkins** pipelines.
- b. Coordinated with developers and testers for release validation and post-deployment support.

5. Documentation & Reporting

- a. Developed and maintained **SOPs, runbooks, and monitoring playbooks**.
- b. Delivered daily system health and performance reports.

This comprehensive monitoring ecosystem ensures **real-time visibility, faster detection, and reliable resolution** of all infrastructure-related issues.

Helpdesk Support

The **Helpdesk team** serves as the **first point of contact** for all end-user issues, covering hardware, software, and access-related support requests.

Key Responsibilities:

- Provide **technical assistance** via phone, email, and ticketing systems.
- Perform **remote troubleshooting** using diagnostic tools.
- Walk users through **problem-solving steps** and escalation when needed.
- Log every call into **Jira**, ensuring full traceability and SLA tracking.
- **Follow up and update** users on ticket progress and resolution.
- Support **Day-One setup** for new hires and manage escalation for unresolved issues.
- Relay **feedback and improvement suggestions** to engineering teams.

Through this structured Helpdesk operation, NuWare has improved **user satisfaction, reduced resolution time**, and maintained consistent service quality across all business functions.

L1 Application Support

In addition to infrastructure and helpdesk operations, NuWare's NOC provides **value-added application support** for ensuring business continuity.

Key Activities:

- Running **application status jobs** through **Jenkins** to confirm service availability.
- Performing **admin tasks** such as start/stop/restart and log cleanup.
- Managing **service restarts** in Cloudera and AWS API Gateway validation.
- Executing **first-level troubleshooting** on application issues and escalating unresolved incidents.

This proactive layer bridges infrastructure monitoring with application reliability, ensuring **end-to-end observability** and minimized downtime.

Performance Metrics

- **P1/P2 Incident Management:** Year-over-year decline in major incidents due to automation and proactive monitoring.
- **Ticket Volume:** 200+ monthly service and incident requests efficiently handled.
- **Assets Managed:** Servers, databases, network devices, and VOIP endpoints under unified monitoring.
- **Response Time:** Under **10 minutes for P1 incidents** and average **<30 minutes for P2** issues.

Outcomes

NuWare's NOC and Helpdesk engagement has led to measurable improvements in performance, reliability, and user experience.

1. Enhanced Uptime and Reliability

- Achieved **99.9% uptime** across applications and infrastructure.
- Prevented downtime through proactive monitoring and automated alert resolution.

2. SLA and Response Excellence

- Maintained **<10-minute response time** for critical alerts.
- Streamlined escalation procedures reduced Mean Time to Resolve (MTTR) by 25%.

3. Operational Efficiency and Automation

- Automated recurring monitoring and reporting tasks through **Jenkins and Ansible**.
- Reduced manual intervention by 30%, freeing resources for optimization initiatives.

4. Improved End-User Experience

- Helpdesk-driven issue resolution improved user satisfaction and onboarding efficiency.
- Consistent communication during incidents improved business confidence.

5. Scalable and Cost-Effective Model

- Flexible offshore support ensured 24x7 coverage at optimized cost.
- Scalable model capable of accommodating future expansion and new tool integrations.